

4.1 Modular Arithmetic

Definition: Congruence

- Let a and b be integers and m a positive integer, then a is **congruent** to b modulo m if m divides $a - b$.

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Example. $m = 3$, then $4 = 1 \bmod 3$

Definition: Congruence

- Let a and b be integers and m a positive integer, then a is **congruent** to b modulo m if m divides $a - b$.

- We write $a \equiv b \pmod{m}$
 $m = 5$, then $4 = 9 = 14 = 19 = \dots = -1 = -6 = \dots$
All these number have remainder 4 when divided by 5.

$$\begin{aligned} 9 &= 5 \times 1 + 4 \\ 14 &= 5 \times 2 + 4 \\ 19 &= 5 \times 3 + 4 \end{aligned}$$

This is called a **congruence** relation, and m is the **modulus**.

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- We write

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- The set Take $m = 5$, $a = 4$, then $5\mathbb{Z} + 4 = \{ \dots, -6, -1, 4, 9, 14, 19, \dots \}$
$$m\mathbb{Z} + a = \{mk + a : k \in \mathbb{Z}\}$$

of all integers congruent to an integer a modulo m is called the **congruence class** of a modulo m .

a : remainder when dividing numbers by m

Easy facts

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$4 \equiv 19 \pmod{5}$$

$$3 \equiv 38 \pmod{5}$$

• $a + c \equiv b + d \pmod{m}$ $4 + 3 = 7$ and $19 + 38 = 57$, then $7 \equiv 57 \pmod{5}$

$a \equiv b \pmod{m}$ means m divides $a - b$, and $c \equiv d \pmod{m}$ means m divides $c - d$,
then $(a+c) - (b+d) = (a-b) + (c-d)$ is a sum of multiples of m , so is the sum.
So m divides $(a+c) - (b+d)$, that means $a + c \equiv b + d \pmod{m}$.

Easy facts

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

- $a + c \equiv b + d \pmod{m}$

- $ac \equiv bd \pmod{m}$ $4 \times 3 = 12$, $19 \times 38 = 722$, check: $12 \equiv 722 \pmod{5}$.

$a \equiv b \pmod{m}$ means m divides $a - b$, and $c \equiv d \pmod{m}$ means m divides $c - d$, then $ac - bd = ac (-bc + bc) - bd = c(a - b) + b(c - d)$ is a linear comb. of multiples of m , so is this linear comb. That means m divides $ac - bd$, and so $ac \equiv bd \pmod{m}$.

Easy facts

If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

- $a + c \equiv b + d \pmod{m}$

- $ac \equiv bd \pmod{m}$

- $a^r \equiv b^r \pmod{m}$ $4 \equiv -1 \pmod{5}$, then (square) $16 \equiv 1 \pmod{5}$,
(cube) $64 \equiv -1 \pmod{5}$, and similar for larger powers.

When $r = 1$, we are already given $a \equiv b \pmod{m}$.

When $r = 2$, take $c = a$ and $d = b$ and use $ac \equiv bd \pmod{m}$, we get $a^2 \equiv b^2 \pmod{m}$.

When $r = 3$, take $c = a^2$ and $d = b^2$, then use $ac \equiv bd \pmod{m}$ again, we get $a^3 \equiv b^3 \pmod{m}$.

For larger r , use $ac \equiv bd \pmod{m}$ again and again...

Example. Find the last two digits of

$$9 + 99 + 999 + 9999 + \dots + \underbrace{99 \dots 9}_{100 \text{ digits of } 9} \quad . \text{ in mod } 100$$

$$\underbrace{9 + 99 + 99 + 99 + \dots + 99}_{99 \text{ many } 99\text{s}} = 9 + 99 \times 99 = 9 + (-1)(-1) = 9 + 1 = 10 \quad \text{mod } 100$$

(Notice that $99 \equiv -1 \pmod{100}$)

Example. Find the last two digits of

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Example. Find the remainder when 2^{100} is divided by 7.

mod $m = 7$

$$2^2 = 4 \bmod 7, \quad 2^3 = 8 = 1 \bmod 7 \quad (\text{lucky})$$

$$2^{3k} = (2^3)^k = 1^k = 1 \bmod 7 \quad \text{for any integer } k. \quad (\text{Any } 3k\text{-power of } 2 \text{ is } 1 \text{ in mod } 7).$$

$$\text{Take } k = 33, \text{ then } 2^{99} = 1 \bmod 7, \text{ then } 2^{100} = 2^1 + 99 = 2 \times 2^{99} = 2 \times 1 = 2 \bmod 7.$$

When $a = 2$ and $m = 7$ are small, so we can do trial-and-error.

When these numbers are huge, then we have to use Fast Modular Exponentiation.

4.2 Integer Representation

$\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ = non-negative integers.

Definition

Let $b \geq 2$ be a positive integer and $m \in \mathbb{N}_0$. Then we call $(\underline{d_k d_{k-1} \dots d_1 d_0})_b$ the **base b expansion** of m if

each d_j is a base- b digit

$$m = \sum_{j=0}^k d_j b^j, \quad \text{and} \quad 0 \leq d_j < b$$

for all $j \in \{0, \dots, k\}$.

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- Ordinary numbers are in base $b = 10$ (decimal expansion).
- Other useful bases include $b = 2$ (binary), $b = 8$ (octal), $b = 16$ (hexadecimal).

Example. What is the decimal expansion of the integer that has 1010111_2 as its binary expansion?

$$\begin{aligned} & 1 \times 2^0 + 1 \times 2^1 + 1 \times 2^2 + 0 \times 2^3 + 1 \times 2^4 + 0 \times 2^5 + 1 \times 2^6 \\ &= 1 + 2 + 4 + 16 + 64 = 87. \end{aligned}$$

Example. What is the decimal expansion of the integer that has 1010111_2 as its binary expansion?

Example. What is the decimal expansion of the number with octal expansion 1276_8 ?

Example. What is the decimal expansion of the number with hexadecimal expansion $2AEB_{16}$?

$$\begin{aligned} 2AEB_{16} &= B + E \times 16 + A \times 16^2 + 2 \times 16^3 \\ &= 11 + 14 \times 16 + 10 \times 16^2 + 2 \times 16^3 \\ &= \end{aligned}$$

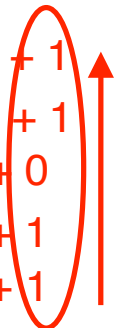
In base 16, the digits are:

0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F,

16_{10} is 10_{16}

Example. Find the binary expansion of 59.

Divide 59 by 2 successively:

$$\begin{aligned} 59 &= 2 \times 29 + 1 \\ 29 &= 2 \times 14 + 1 \\ 14 &= 2 \times 7 + 0 \\ 7 &= 2 \times 3 + 1 \\ 3 &= 2 \times \boxed{1} + 1 \end{aligned}$$


$$59 = \underline{1\ 1\ 1\ 0\ 1\ 1}_2$$

Example. Find the binary expansion of 59.

Example. Find the octal expansion of 2020.

Example. Find the hexadecimal expansion of 2020.

$$\begin{aligned} 2020 &= 16 \times 126 + 4 \\ 126 &= 16 \times 7 + 14 \\ 7 &< 16 \end{aligned}$$

$$2020 = 7E4_{16}$$

Fast modular exponentiation (taking powers)

If we want to calculate $a^n \pmod{m}$:

1. Find the binary expansion (base $b = 2$) of n .
2. Calculate

$$a, a^2, a^4, a^8, a^{16}, \dots, a^{2^k} \pmod{m}$$

by taking successive squares.

3. Multiply the necessary powers of a .

Example. Find $3^{21} \pmod{29}$. Step 1: the power $21 = \underline{16} + 4 + 1$

Step 2: $a = 3$, $a^2 = 9$, $a^4 = (a^2)^2 = 9^2 = 81 = 23 = -6 \pmod{29}$

$$a^8 = (a^4)^2 = (-6)^2 = 36 = 7 \pmod{29}, \quad a^{16} = (a^8)^2 = 7^2 = 49 = -9 \pmod{29}$$

$$\text{Step 3: } 3^{21} = 3^{16+4+1} = 3^{16} \times 3^4 \times 3^1 = (-9)(-6)3 = 54 \times 3 = (-4) \times 3 = -12 = 17 \pmod{29}$$